JABstem

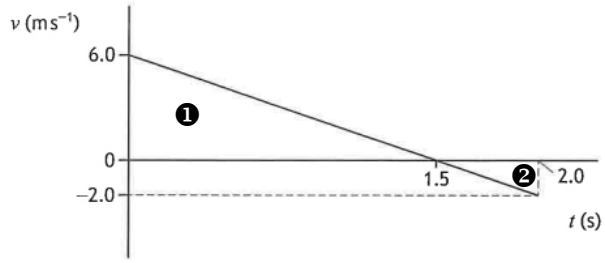
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Past Papers Nat 5 Physics 2025 Marking Scheme

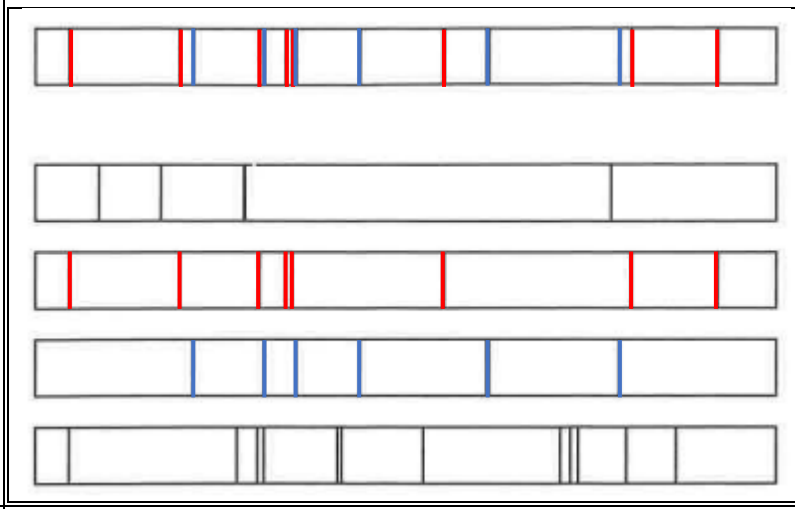
Grade Awarded	Mark Required		% candidates achieving grade
	/125	%	
A	84+	67%+	32.8%
B	70+	56%+	20.7%
C	56+	45%+	19.1%
D	42+	34%+	15.3%
No award	<42	<34%	12.1%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	15.6 /25	38.1 /75	17.3 /25

2025 Nat5 Physics Marking Scheme

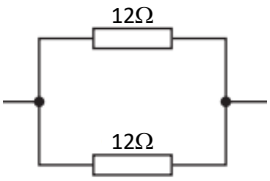
<table>
<tr><th>Question</th><th>Answer</th><th>% Correct</th><th>Physics Covered</th></tr>
<tr><td>1</td><td>B</td><td>74</td><td>
<table>
<tr><td>Scalar Quantity</td><td>energy</td><td>speed</td><td>distance</td><td>time</td><td>mass</td></tr>
<tr><td>Vector Quantity</td><td>force</td><td>velocity</td><td>displacement</td><td>acceleration</td><td>weight</td></tr>
</table>
</td></tr>
<tr><td>2</td><td>A</td><td>42</td><td>


<table>
<tr><td>Area ❶</td><td>Area ❷</td><td rowspan="4">Total displacement = 4.5m – 0.5m = 4.0m</td></tr>
<tr><td>Distance = area under graph</td><td>Distance = area under graph</td></tr>
<tr><td>= ½ x 1.5 x 6.0</td><td>= ½ x 0.5 x -2</td></tr>
<tr><td>= 4.5m</td><td>= -0.5m</td></tr>
</table>
</td></tr>
<tr><td>3</td><td>D</td><td>59</td><td>
a = -2 m s^{-2} v = 5 m s^{-1} u =? t = 8.0 s
$a = \frac{v - u}{t}$
$-2 = \frac{5 - u}{8}$
$-16 = 5 - u$
$u = 5 + 16$
$u = 21 \text{ m s}^{-1}$
</td></tr>
<tr><td>4</td><td>A</td><td>65</td><td>
☑A Force of the table on the cup is balanced by the force of cup on the table
☒B Force of the cup on the Earth is balanced by the force of Earth on the cup
☒C Force of the Earth on cup is balanced by force of the cup on Earth
☒D Force of the cup on the table is balanced by force of the table un the cup
☒E Force of the table on the Earth is balanced by the force of Earth on the table
</td></tr>
<tr><td>5</td><td>E</td><td>63</td><td>
W = ? m = 180kg g = 9.8 N kg^{-1}
W = m g
W = 180 x 9.8
F = 1764 N
E_w = ? F = 1764 N d = 3.0m
E_w = F d
E_w = 1764 x 3.0
F = 5292 J
</td></tr>
<tr><td>6</td><td>D</td><td>57</td><td>
E_k = ? m = 25kg v = 9200 m s^{-1}
$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 25 \times (9200)^2 = 1.058 \times 10^9 \text{ J} = 1.1 \times 10^9 \text{ J}$
</td></tr>
<tr><td>7</td><td>C</td><td>62</td><td>
v = ? r = 6.4x10^6 + 3.6x10^7 = 4.24x10^7 m g = 0.22 N kg^{-1}
$v = \sqrt{rg} = \sqrt{4.24 \times 10^7 \times 0.22} = \sqrt{9.33 \times 10^6} = 3054 \text{ m s}^{-1} = 3100 \text{ m s}^{-1}$
</td></tr>
</table>

8	C	76	W = ? m = 3.5kg g = 7.7 N kg^{-1} W = m x g W = 3.5 x 7.7 W = 27 N
9	E	62	d = ν x t d = 3.0×10^8 x 8.6 x 365.25 x 24 x 60 x 60 d = 8.1×10^{16} m
10	C	91	 star hydrogen helium carbon nitrogen
11	E	87	**Trace X — d.c. signal** • negative charges (electrons) flow in one direction only. • gives a constant trace on oscilloscope **Trace Y — d.c. signal** • negative charges (electrons) flow in opposite direction to X • gives a constant trace on oscilloscope **Trace Z — a.c. signal** • the direction of electrons in current changes back and forth at regular intervals • the size of the current varies with time and is not constant
12	A	67	☑A Q+R must have same charge but opposite charge to particle ☒B Q+R must have same charge as particle is attracted to both Q+R ☒C Q+R must have same charge as particle is attracted to both Q+R ☒D Q+R must have same charge as particle is attracted to both Q+R ☒E particle must have opposite charge to Q+R for particle to be attracted to Q+R
13	D	50	Voltage in 240Ω resistor = 3.0V ∴ Voltage in unknown resistor = 12.0V - 3.0V = 9.0V V_1 = 3.0V V_2 = 9.0V R_1 = 240V R_2 = ? $\frac{V_1}{V_2} = \frac{R_1}{R_2}$ $\frac{3.0}{9.0} = \frac{240}{R_2}$ $R_2 \times 3.0 = 240 \times 9.0$ $R_2 = \frac{240 \times 9.0}{3.0}$ $R_2 = 720\ \Omega$

14 D 65

Combining Parallel Resistors:



$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ (1 mark)

$\frac{1}{R_T} = \frac{1}{12} + \frac{1}{12}$ (1 mark)

$\frac{1}{R_T} = \frac{2}{12}$

$R_T = {}^{12}/_{2} = 6\ \Omega$

Combining Series Resistors:



$R_T =$ 12Ω + 6Ω = **18 Ω**

15 A 64

Voltage dissipated in lamp

V = ? I = 0.5 A R = 8.0 Ω

V = I R

V = 0.5 x 8.0

V = 4.0 V

P = ? V = 4.0 V R = 8.0 Ω

$P = \frac{V^2}{R} = \frac{(4)^2}{8.0} = \frac{16}{8.0} = 2.0$ W

16 B 65

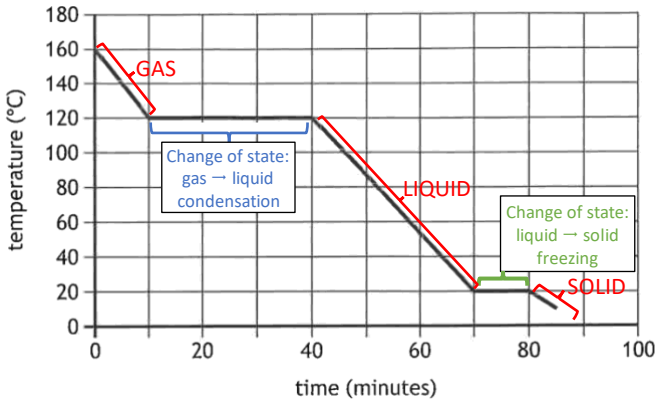
Kettle = 2.6kW = 2600W	Television = 170W	Microwave = 900W
$I = \frac{P}{V} = \frac{2600}{230} = 11.30A$	$I = \frac{P}{V} = \frac{170}{230} = 0.74A$	$I = \frac{P}{V} = \frac{900}{230} = 3.91A$
13A fuse required. Power rating is above 720W	3A fuse required. Power rating is below 720W	13A fuse required. Power rating is above 720W

17 C 34

Statement I - Correct	Statement II - Correct	Statement III - Incorrect
Insulation of the beaker would reduce the heat loss to the surroundings by trapping a layer of air around the beaker	Moving the immersion heater further into the water should increase the heat transferred into the water from the heater.	The joulemeter will directly measure the quantity of energy transferred to the water without the need to calculate E from E = P t

18 C 54

Statement I - Correct	Statement II - Incorrect	Statement III - Correct
The substance evaporates/boils and condenses at 120°C. The temperature does not change during a change of state.	The conversion of gas to liquid is complete at 40 minutes so it will be 100% liquid at 40 minutes	The substance freezes and melts at 120°C. The temperature does not change during a change of state.



19 D 31

P = ? F = 520N A = $1.6\text{x}10^{-3}$ m² per shoe ∴ $3.2\text{x}10^{-3}$ m²

$$P = \frac{F}{A} = \frac{520}{3.2\text{x}10^{-3}} = 1.6\text{x}10^{5}\ \text{Pa}$$

20 D 41

Volume: Constant $p_1 = 1.90\times10^5$ Pa $T_1 = 20^oC = 293$ K; $p_2 = ?$ $T_2 = 35^oC = 308$ K

$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$

$$\frac{1.90\times10^5}{p_2} = \frac{293}{308}$$

$$p_2 \times 293 = 1.90\times10^5 \times 308$$

$$p_2 = \frac{1.90\times10^5 \times 308}{293}$$

$$p_2 = 2.0\times10^5 \text{ Pa}$$

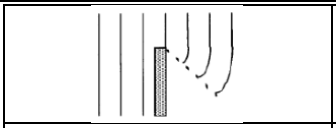
21 B 83

Wavelength	Amplitude
3 waves = 120mm 1 wave = $^{120}/_{3}$ = 40mm	Amplitude = $\frac{0.2}{2}$ = 0.1m

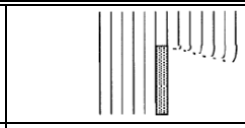
22 B 56

☒A gamma rays are shorter than visible light and diffract less than visible light
☑B infrared diffracts more than visible as it is longer and has greater frequency
☒C ultraviolet rays are shorter than visible light and diffract less than visible light
☒D radio waves have a lower frequency than microwave radiation
☒E X-rays rays are shorter than visible light and diffract less than visible light

EM Type	Gamma	X-Ray	Ultra-violet	Visible	Infra-Red	Microwave	Radio & TV
Energy	High ←						→ Low
Frequency	High ←						→ Low
Wavelength	Low ←						→ High



Long Wave diffraction



Short Wave diffraction

Wavelength must be same before *and* after obstacle.
Longer waves diffract more than shorter waves

23 E 49

A = 5.2 MBq = 5.2×10^6 Bq N = ? t = 1.2 hours = 1.2 x 60 x 60 s

$$A = \frac{N}{t} \quad \therefore N = A \quad t$$

$$N = 5.2\times10^6 \times 1.2 \times 60 \times 60$$

$$N = 2.2\times10^{10}$$

24 A 69

$\dot{H}$ = 6.0 μSv h^{-1} H = 480 μSv t = 5 minutes = 5x60 s

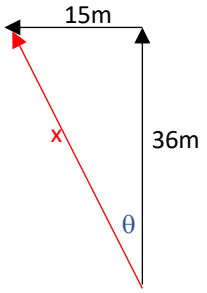
$$\dot{H} = \frac{H}{t} \quad \text{(1 mark)}$$

$$\dot{H} = \frac{480}{5\times60} \quad \text{(1 mark)}$$

$$\dot{H} = 1.6 \text{ μSv } s^{-1} \quad \text{(1 mark)}$$

25 B 87

☒A Fission: large nucleus splits into two or more smaller nuclei
☑B Fusion: two nuclei of smaller mass combine to produce a nucleus of larger mass
☒C Alpha decay: helium nucleus emitted from a nucleus
☒D Beta Decay: fast electron emitted from a nucleus
☒E Gamma Decay: electromagnetic radiation emitted from a nucleus

Question	Answer	Physics Covered
1a(i)	39m	Horizontal displacement = 15m (WEST or 270) Vertical displacement = 32m + 4m = 36m (North or 000)  $x = \sqrt{(36)^2 + (15)^2}$ $x = \sqrt{1296 + 225}$ $x = \sqrt{1521}$ $x = 39\text{m}$
1a(ii)	337	$\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{15}{36} = 0.417 \quad \therefore \ \theta = 22.6^o$ Bearing = 360° – 23° = 337°
1b	0.71 m s^{-1} at 337	d = 39 m $\bar{v}$ = ? t = 55 s d = $\bar{v}$ t (1 mark) 39 = $\bar{v}$ x 55 (1 mark) $\bar{v}$ = $^{39}/_{55}$ = 0.71 m s^{-1} (1 mark)
1c	3500 J	E_w = ? F = 68N d = 32+15+4 = E_w = F d (1 mark) E_w = 68 x 51 (1 mark) E_w = 3468 N (1 mark)
2a(i)	1.40x10^7 N	W = ? m = 1.43x10^6 kg g = 9.8 N kg^{-1} W = m x g (1 mark) W = 1.43x10^6 x 9.8 (1 mark) W = 1.40x10^7 N (1 mark)
2a(ii)	6.15 m s^{-2}	a = ? F = 2.28x10^7 – 1.40x10^7 = 0.88X10^7 N m = 1.43x10^6 kg $a = \frac{F}{m} = \frac{0.88\text{x}10^7}{1.43\text{x}10^6} = 6.15\ \text{m s}^{-2}$ (1 mark) (1 mark) (1 mark)
2b(i)	Diagram showing:	Direction Up ↑ One from: Friction Air resistance (1 mark)  (1 mark) Direction Down ↓ One from: weight pull of gravity gravitational pull force due to gravity'
2b(ii)	Answer to Include:	1st mark: air resistance/friction/upwards force/drag *increases* 2nd mark: produces a new unbalanced force upwards
2b(iii)	Answer to Include:	1st mark: upward force = 9300N 2nd mark: Constant velocity (means that forces must be balanced/equal but oppsitie.

3

Open ended question:

0 marks	1 mark	2 marks	3 marks
16%	48%	27%	9%

1 mark	2 marks	3 marks
Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.

4a

Answer to include:

1st mark	It experiences a (gravitational) slingshot/catapult (from Mars)
2nd mark	(causing) an increase in its speed/velocity/kinetic energy.

4b(i)

Answer to include:

Small unbalanced force will produce a small acceleration which will continuously increase the velocity over extended/long period of time.

4b(ii)

Answer to include:

The further the distance from the sun the less power produced by solar cells.

4c(i)

1.4 days

41 orbits ⟷ 56 days

1 orbit ⟷ 56 days $\times \frac{1}{41}$

= 1.37 days

4c(ii)

Orbital period decreases close to Psyche

The closer an object is to the object it is orbital, the lower the period is as it has to travel a higher velocity to stay in that orbit.

5a

The positive charges in metal dish **or** has same charge as dome **or** like/same charges repel

As electrons are withdrawn from the drawn to create the positive charge in the dome, the foil dishes becoming increasing positively charged. The dishes repel each other and one by one the dishes are pushed upwards and fall.

5b(i)

3.1×10^{-3} A

$Q = 2.50 \times 10^{-6}$ C $I = ?$ $t = 0.80$ ms $= 0.8 \times 10^{-3}$ s

$Q = I \quad t$ (1 mark)

$2.50 \times 10^{-6} = I \times 0.8 \times 10^{-3}$ (1 mark)

$I = \frac{2.50 \times 10^{-6}}{0.8 \times 10^{-3}} = 3.13 \times 10^{-3}$ A (1 mark)

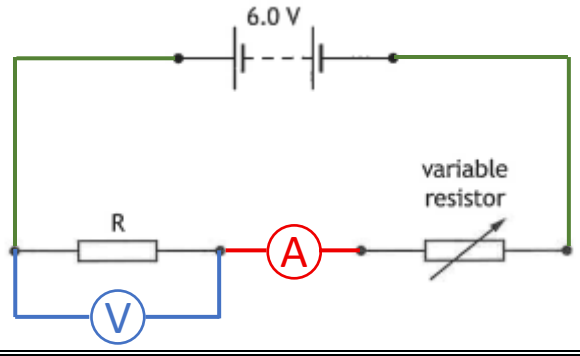
5b(ii)

1.56×10^{13}

$$\text{number of electrons} = \frac{\text{discharge}}{\text{charge on 1 electron}} = \frac{2.50 \times 10^{-6}}{1.60 \times 10^{-19}} = 1.56 \times 10^{13} \text{ electrons}$$

6a

Diagram showing:



1 mark:
Ammeter connected in series.
Ammeter can be moved to any point in circuit diagram represented by green connecting leads.

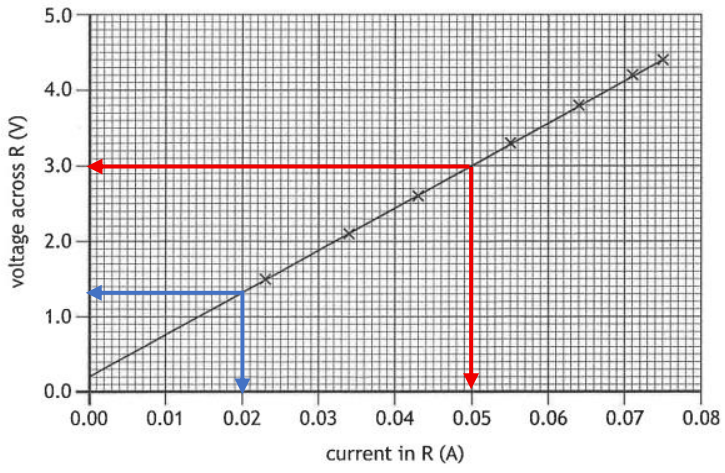
1 mark:
Voltmeter must be connected in parallel around the resistor R.
No other possible position for voltmeter.

Connecting leads must complete the circuit to be correct

6b

56.7 Ω

Range Accepted
54.5-57.5



From graph:
When current = 0.02 A then Voltage = 1.3V
When current = 0.05 A then Voltage = 3.0V

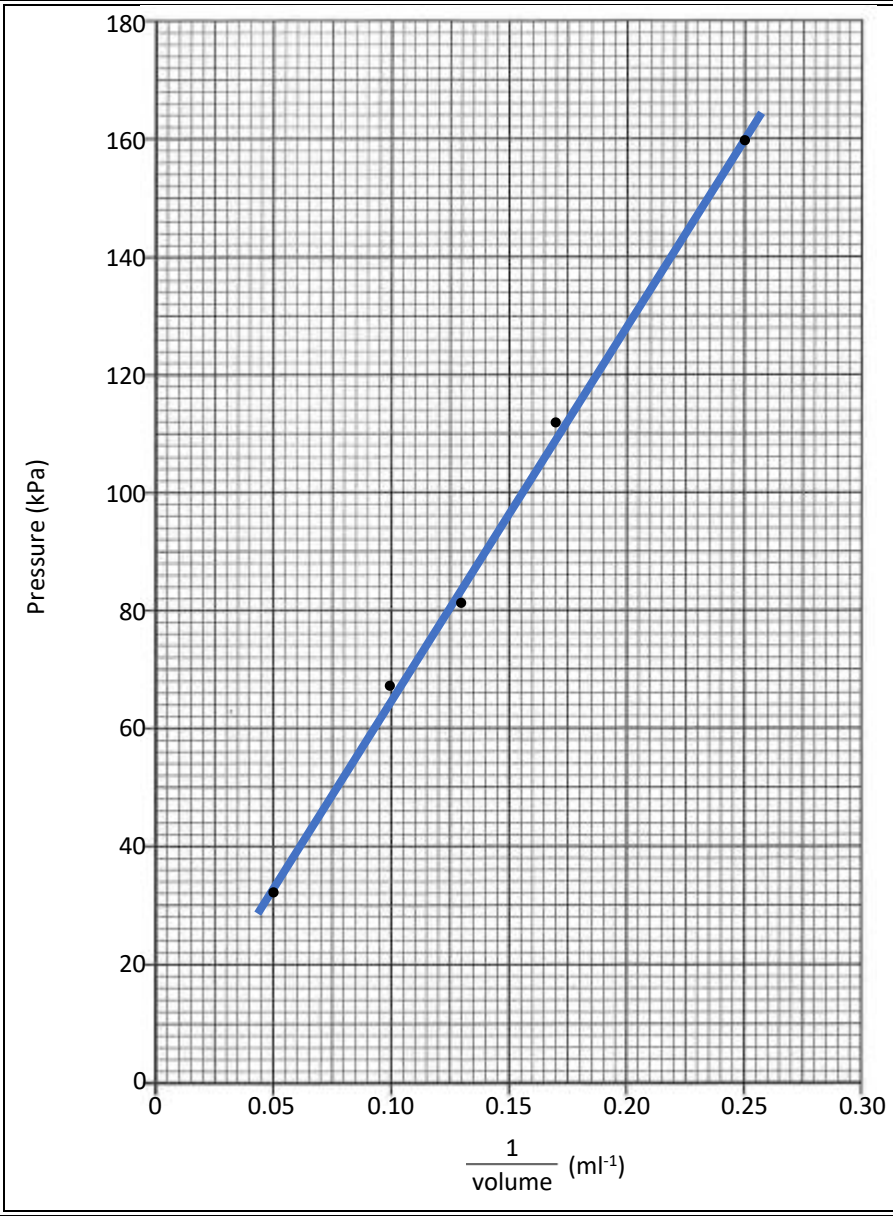
$R = \frac{y_2 - y_1}{x_2 - x_1}$

$R = \frac{3.0 - 1.3}{0.05 - 0.02}$

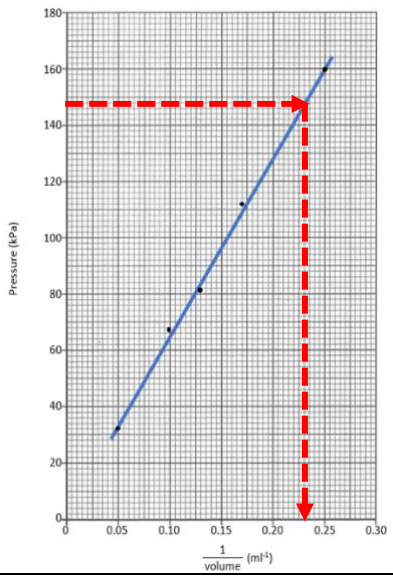
$R = \frac{1.7}{0.03}$

$R = 56.7\ \Omega$

<table>
<tr><td>6c</td><td>Resistance of lamp is changing/increasing</td><td>As the temperature of the lamp in increasing, the resistance of the filament increases.
The resistance of the lamp must be increasing as the gradient of the line on the is increasing.
The gradient of the line is the resistance due to the equation $R = {}^{V}/_{I}$.</td></tr>
<tr><td>7a</td><td>One spotlight goes out rest stay on</td><td>The LED spotlights are arranged in a parallel circuit so that if one spotlight is broken then the remaining spotlight stay working as it is connected in parallel.</td></tr>
<tr><td>7b</td><td>1.6 A</td><td>P = 4.8 x 4 = 19.2W V = 12 V I = ?
$P = I \quad V$ (1 mark)
$19.2 = I \times 12$ (1 mark)
$I = \frac{19.2}{12} = 1.6\ A$ (1 mark)</td></tr>
<tr><td>7c(i)</td><td>Mosfet or transistor</td><td></td></tr>
<tr><td>7c(ii)</td><td>Answer to include:</td><td>1st mark (when light decreases) resistance in LDR increases
2nd mark When voltage across the LDR increases
3rd mark (when voltage across LDR reaches a certain value) Mosfet switches on</td></tr>
<tr><td>8</td><td>Open ended question:
0 marks 39%; 1 mark 37%; 2 marks 18%; 3 marks 7%</td><td>1 mark: Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.
2 marks: Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.
3 marks: Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.</td></tr>
<tr><td>9a(i)</td><td>35000 J</td><td>E_h = ? c = 4180 J kg^{-1} ^{o}C^{-1} m = 0.38 kg ΔT= 22°C
$E = c \times m \times \Delta T$ (1 mark)
$E = 4180 \times 0.38 \times 22$ (1 mark)
$E = 34945\ J$ (1 mark)</td></tr>
<tr><td>9a(ii)</td><td>290 s</td><td>P = 120 W E = 35000 J t = ?
$P = \frac{E}{t}$ (1 mark)
$120 = \frac{35000}{t}$ (1 mark)
$t = \frac{35000}{120} = 290\ s$ (1 mark)</td></tr>
<tr><td>9a(iii)</td><td>One example from:</td><td>Insulate the machine better | Put a lid on</td></tr>
<tr><td>9b</td><td>0.0458kg</td><td>E = 15.3kJ = 15300J m = ? l = 3.34x10^{-5} J kg^{-1}
$E = m \times l$ (1 mark)
$15300 = m \times 3.34\times10^{-5}$ (1 mark)
$m = \frac{15300}{3.34\times10^{-5}} = 0.0458\ kg$ (1 mark)</td></tr>
<tr><td>10a</td><td>Particles hit the walls of the syringe</td><td>Pressure is caused by the particles of gas in air hitting the walls of the container. The greater the pressure the greater the number of collisions.</td></tr>
</table>

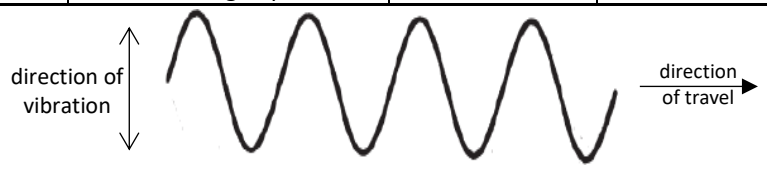
<table>
<tr>
<td>10b(i)</td>
<td>Line graph as shown</td>
<td> 
</td>
<td><u>1 mark</u>
suitable scales, labels and units

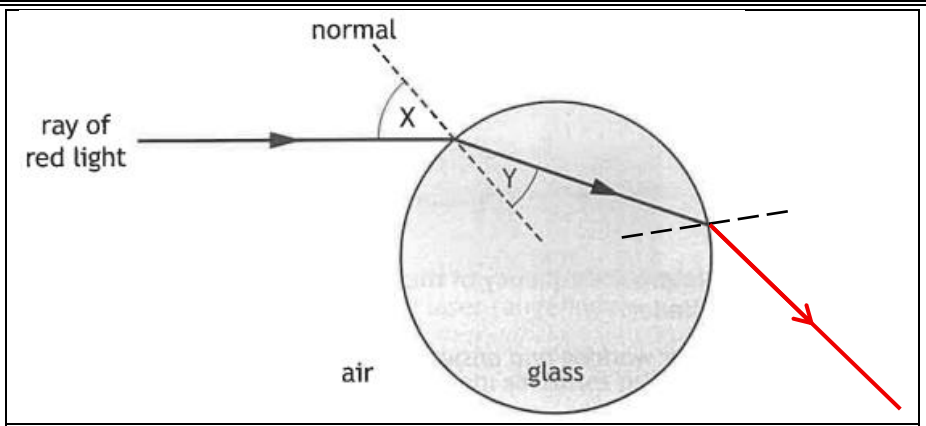
<u>1 mark</u>
all points plotted accurately to ± half a division

<u>1 mark</u>
best fit straight line</td>
</tr>
<tr>
<td>10b(ii)</td>
<td>Pressure is inversely proportional to volume</td>
<td colspan="2">The graph shows a directly proportional relationship between $^{1}/_{volume}$ and the pressure.
This means the volume is inversely proportional to the pressure which results in an increase in volume giving a decrease in pressure and vice versa.</td>
</tr>
<tr>
<td>10b(iii)</td>
<td>4.3ml</td>
<td>  </td>
<td>when pressure = 148 kPa

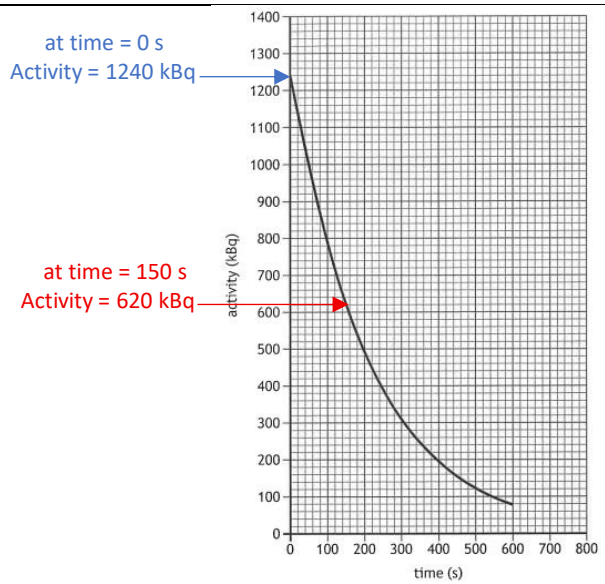
then $^{1}/_{volume}$ = 0.23 ml^{-1}

then volume = $^{1}/_{0.23ml^{-1}}$ = 4.3ml</td>
</tr>
</table>

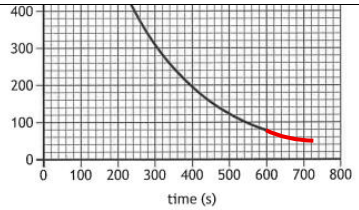
10c	One answer from:	Repeat (measurements) and average. / Repeat (measurements) to identify outliers/rogue points. / Increase the range of volumes. / Increase the number of different volumes
11a	Direction of vibration at right angles to the direction of wave travel.	**Transverse Waves** 
11b(i)	Working showing:	N = 500 t = 30minutes = 30x60 s f =? $f = \frac{N}{t} = \frac{500}{30x60} = 0.28\ Hz$ (1 mark) (1 mark)
11b(ii)	5.0 m s^{-1}	d = 160 m v = ? t =32 s d = v x t (1 mark) 160 = v x 0.32 (1 mark) v = $^{160}/_{0.32}$ = 5.0 m s^{-1} (1 mark)
11b(iii)	18.0 m	v = 5.0 m s^{-1} f = 0.278 Hz λ = ? v = f x λ (1 mark) 5.0 = 0.278 x λ (1 mark) λ = $^{5.0}/_{0.278}$ = 18.0 m (1 mark)
12a(i)	0.067 s	d = 20200 km = 2.02x10^7m v = 3.0x10^8 m s^{-1} t =? d = v x t (1 mark) 2.02x10^7 = 3.0x10^8 x t (1 mark) $t = \frac{2.02x10^7}{3.0x10^8}$ t = 0.067s (1 mark)
12a(ii)	Answer to include:	1 mark: Not geostationary 1 mark: It does not have an (orbital) period of 24 hours **Or** It is not at an (orbital) altitude of 36000km
12b(i)	one answer from:	CCD / Photodiode / phototransistor
12b(ii)	3.3x10^{14} Hz	v = 3.0x10^8 m s^{-1} f = ? λ = 904 nm = 904x10^{-9} m v = f x λ (1 mark) 3.0x10^8 = f x 904x10^{-9} (1 mark) $f = \frac{3.0x10^8}{904x10^{-9}}$ f = 3.32x10^{14} Hz (1 mark)
12b(iii)	180 m	d = ? v = 3.0x10^8 m s^{-1} t = 1.2x10^{-6} s Total d = v x t (1 mark) Total d = 3.0x10^8 x 1.2x10^{-6} (1 mark) Total d = 360 m (1 mark) d = 180m (1 mark)
13a	Answer to include:	X: Angle of incidence / Y: Angle of refraction

<table>
<tr><td>13b</td><td>Answer to include:</td><td>
<table>
<tr><td>1 mark</td><td>1 from:
<table>
<tr><td>Refraction takes place</td><td>There is a change in speed.</td><td>The ray is moving from a less (optically) dense into a more (optically) dense medium.</td></tr>
</table></td></tr>
<tr><td>1 mark</td><td>The angle of incidence is greater than 0º</td></tr>
</table>
</td></tr>
<tr><td>13c</td><td>Diagram showing:</td><td>


Ray emerging from glass changes direction to an angle further from the normal
</td></tr>
<tr><td>14a</td><td>Answer to include:</td><td>Alpha and Beta radiation will not travel through the various layers to the surface to be detected</td></tr>
<tr><td>14b(i)</td><td>Time taken for activity to decrease by half</td><td>Half-life is defined as the time taken for the activity/corrected count rate of a radioactive source to half.</td></tr>
<tr><td>14b(ii)</td><td>1 mark:
Sodium-24

1 mark:
Gamma emitter and suitable halflife</td><td>
<table>
<tr><th>Source</th><th>Half-Life</th><th>Radiation emitted</th><th>Reasoning</th></tr>
<tr><td>sodium-24</td><td>15 hours</td><td>beta & gamma</td><td>Source chosen as half-life is not to short or too long and releases gamma radiation which can be detected on the surface</td></tr>
<tr><td>bismuth-204</td><td>11.2 hours</td><td>beta</td><td>Radiation would be stopped before the surface and not detected from the surface.</td></tr>
<tr><td>barium-133</td><td>10.5 years</td><td>gamma</td><td>Half-life is too long and would be present in the environment for many years</td></tr>
<tr><td>barium-137m</td><td>2.6 minutes</td><td>gamma</td><td>Half-life is too short as the radioactivity would decrease too much before all of pipe hade been examined</td></tr>
</table>
</td></tr>
<tr><td>14c(i)</td><td>150 s</td><td>


</td></tr>
<tr><td>14c(ii)</td><td>50 kBq

Accept 40-60 kBq</td><td>


</td></tr>
<tr><td>15a(i)</td><td>2.5x10^{-7} Gy</td><td>
H = 0.25 µSv = 0.25x10^{-6} Sv D = ? W_r = 1
H = D x W_r (1 mark)
0.25x10^{-6} = D x 1 (1 mark)
D = 0.25x10^{-6} = 2.5x10^{-7} Gy (1 mark)
</td></tr>
</table>

<table>
<tr><td>15a(ii)</td><td>1.6×10^{-5} J
or 0.000016 J</td><td>D = 2.5×10^{-7} Gy E = ? m = 64 kg
$D = \frac{E}{m}$ ∴ $2.5\times10^{-7} = \frac{E}{64}$ ∴ E = $2.5\times10^{-7} \times 64 = 1.6\times10^{-5}$ J
(1 mark) (1 mark) (1 mark)</td></tr>
<tr><td>15b</td><td>Answer to include:</td><td>Process where atoms become charged by losing or gaining electrons</td></tr>
<tr><td>15c</td><td>One answer from:</td><td><table><tr><td>Stand behind screen</td><td>Limit time near machine</td><td>Keep distance from machine</td><td></td></tr></table></td></tr>
</table>

2025 National 5 Physics Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	E	Missing	Invalid
1	B	9	**74**	10	3	3	1	0
2	A	**42**	8	12	34	2	2	0
3	D	4	23	9	**59**	3	1	0
4	A	**65**	2	5	12	14	1	0
5	E	2	23	7	3	**63**	2	0
6	D	7	26	5	**57**	3	2	0
7	C	18	13	**62**	5	2	1	0
8	C	2	8	**76**	4	8	1	0
9	E	20	5	4	8	**62**	1	0
10	C	1	2	**91**	4	1	1	0
11	E	1	3	7	1	**87**	1	0
12	A	**67**	8	7	10	7	1	0
13	D	7	18	7	**50**	15	2	0
14	D	8	5	3	**65**	18	1	0
15	A	**64**	15	13	4	3	1	0
16	B	3	**65**	4	12	15	1	0
17	C	12	1	**34**	27	24	1	0
18	C	17	8	**54**	8	11	1	0
19	D	6	3	8	**31**	51	1	0
20	D	5	5	4	**41**	44	2	0
21	B	2	**83**	8	3	3	1	0
22	B	7	**56**	10	21	4	1	0
23	E	14	9	15	12	**49**	2	0
24	A	**69**	10	6	7	7	2	0
25	B	10	**87**	1	1	1	1	0

2025 National 5 Physics Section 2 Marks Data Analysis

2025 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	2	1.1	56.4%	52	3.8%
Q1aii	2	0.7	37.0%	60	9.0%
Q1b	3	1.5	48.9%	53	4.1%
Q1c	3	2	65.5%	46	6.8%
Q2ai	3	2.5	81.7%	56	1.0%
Q2aii	4	2	50.6%	67	5.6%
Q2bi	2	1	51.3%	56	2.7%
Q2bii	2	0.6	29.2%	46	3.1%
Q2biii	2	0.7	34.5%	56	21.7%
Q3	3	1.3	43.3%	54	6.0%
Q4a	2	0.9	46.8%	42	4.8%
Q4bi	1	0.1	5.7%	19	7.2%
Q4bii	1	0.4	36.7%	33	6.4%
Q4ci	1	0.3	33.5%	39	7.3%
Q4cii	1	0.7	68.5%	41	5.8%
Q5a	1	0.6	56.2%	32	5.3%
Q5bi	3	2.1	70.8%	63	4.7%
Q5bii	1	0.4	36.0%	54	19.2%
Q6a	2	1.5	73.1%	53	2.8%
Q6b	2	0.4	20.5%	39	9.3%
Q6c	1	0.3	34.4%	32	6.0%
Q7a	1	0.3	31.5%	28	3.7%
Q7b	3	1.6	54.2%	46	5.4%
Q7ci	1	0.7	65.9%	50	9.1%
Q7cii	3	0.8	27.3%	64	10.8%
Q8	3	0.9	30.8%	60	11.4%
Q9ai	3	1.9	62.0%	64	5.3%
Q9aii	3	2.1	71.3%	63	9.5%
Q9aiii	1	0.5	54.6%	31	6.7%
Q9b	3	1.2	40.3%	62	14.0%
Q10a	1	0.3	31.4%	44	16.3%
Q10bi	3	1.8	59.2%	48	2.3%
Q10bii	1	0.1	7.2%	31	8.1%
Q10biii	2	0.6	28.2%	44	9.4%
Q10c	1	0.4	38.5%	36	8.9%
Q11a	1	0.3	26.1%	47	7.1%
Q11bi	2	1.2	61.2%	48	4.7%
Q11bii	3	2.5	81.8%	45	3.4%
Q11biii	3	2.3	75.3%	60	7.5%
Q12ai	2	1.2	60.3%	70	12.9%
Q12aii	2	1.2	57.7%	55	8.0%
Q12bi	1	0.2	18.8%	33	19.4%
Q12bii	3	1.6	54.3%	71	13.3%
Q12biii	4	2.2	54.2%	72	16.0%
Q13a	1	0.7	69.3%	49	6.4%
Q13b	2	0.7	33.7%	46	8.0%
Q13c	1	0.4	39.8%	30	5.1%
Q14a	1	0.4	40.3%	32	5.4%
Q14bi	1	0.5	54.1%	50	4.5%
Q14bii	2	0.7	36.7%	39	4.8%
Q14ci	1	0.6	62.1%	50	7.5%
Q14cii	1	0.6	62.5%	21	5.5%
Q15ai	3	1.5	49.3%	72	9.0%
Q15aii	3	1.8	59.8%	66	19.5%
Q15b	1	0.4	42.0%	47	12.3%
Q15c	1	0.6	59.3%	27	7.9%
Paper	110	55.7	50.7%		

2025 National 5 Physics Section 2 Frequency of Marks Analysis

2025 Item	0	1	2	3	4
Q1ai	41	4	54		
Q1aii	52	22	26		
Q1b	35	4	41	20	
Q1c	11	32	5	52	
Q2ai	2	2	44	52	
Q2aii	15	42	1	6	35
Q2bi	32	33	35		
Q2bii	48	46	6		
Q2biii	61	9	30		
Q3	16	48	27	9	
Q4a	36	34	30		
Q4bi	94	6			
Q4bii	63	37			
Q4ci	66	34			
Q4cii	31	69			
Q5a	44	56			
Q5bi	10	13	32	45	
Q5bii	64	36			
Q6a	16	21	62		
Q6b	77	4	18		
Q6c	66	34			
Q7a	69	31			
Q7b	13	48	2	37	
Q7ci	34	66			
Q7cii	55	21	11	13	
Q8	39	37	18	7	
Q9ai	17	15	33	35	
Q9aii	16	9	22	54	
Q9aiii	45	55			
Q9b	38	20	24	17	
Q10a	69	31			
Q10bi	20	16	29	35	
Q10bii	93	7			
Q10biii	52	40	8		
Q10c	62	38			
Q11a	74	26			
Q11bi	15	48	37		
Q11bii	12	6	6	76	
Q11biii	14	8	14	63	
Q12ai	25	30	45		
Q12aii	39	7	54		
Q12bi	81	19			
Q12bii	27	14	28	31	
Q12biii	21	12	31	1	35
Q13a	31	69			
Q13b	35	63	2		
Q13c	60	40			
Q14a	60	40			
Q14bi	46	54			
Q14bii	58	10	32		
Q14ci	38	62			
Q14cii	38	62			
Q15ai	35	16	17	33	
Q15aii	30	8	14	48	
Q15b	58	42			
Q15c	41	59			

2025 National 5 Physics Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.9	87.2%	24
2	3	1.5	51.1%	43
3a	1	0.6	58.6%	28
3b	1	0.9	92.0%	16
3c	1	0.7	67.2%	31
3d	1	0.6	62.1%	29
3e	1	0.9	92.8%	29
3f	1	0.9	86.7%	28
4a	1	1	96.0%	21
4b	1	0.9	89.0%	26
4c	1	0.9	89.6%	27
4d	1	0.3	30.1%	27
5	1	0.6	55.2%	37
6	1	0.6	57.2%	32
7	3	0.6	30.6%	23
8a	1	1	97.3%	13
8b	1	1	98.5%	19
Paper	20	13.7	68.7%	

Item	Max Mark	0	1	2	3
1	1	13	87		
2	3	10	42	33	15
3a	1	41	59		
3b	1	8	92		
3c	1	33	67		
3d	1	38	62		
3e	1	7	93		
3f	1	13	87		
4a	1	4	96		
4b	1	11	89		
4c	1	10	90		
4d	1	70	30		
5	1	45	55		
6	1	43	57		
7	2	65	9	26	
8a	1	3	97		
8b	1	1	99		
Paper	20				

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation.
2	3	An account of physics relevant to the aim of the investigation.
3a	1	A brief description of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in a correctly produced table
3d	1	Mean and/or derived values calculated correctly
3e	1	Data relevant to the experiment from an internet/literature source
3f	1	A reference for the source of the internet/literature data
4a	1	An appropriate format from the options of scatter, line or bar graph.
4b	1	The axis/axes of the graph has/have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit
5	1	Valid comparison of the experimental data with data from internet/literature source.
6	1	A conclusion that relates to aim and supported by all the data in the report.
7	2	An evaluation of the experimental procedure.
8a	1	An informative title.
8b	1	A clear and concise report.